The empirical recurrence for OEIS A275223, proved by counting patterns

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Abstract

OEIS A275223 counts arrays in which no entry may repeat a value found at any of three stated offsets, and which are moreover written in canonical form ("new values introduced in order 0..2"). The canonical-form clause is not local and no transfer matrix over the alphabet sees it; but the condition is stated in equalities, so what is counted is equality patterns, and the pattern counts are recovered from the counts L_i over an i-letter alphabet by inverting a falling factorial: $3!\,a(n) = \sum_i \binom{3}{i} D_{3-i} L_i(n)$ with D_m the derangement numbers. Some of the offsets reach two rows back, so each L_i is a walk count whose states are ordered pairs of consecutive rows; relabelling-invariance makes that chain lumpable, cutting 794 pairs to $S = 155$ states. The entry's empirical recurrence of order 9, contributed by R. H. Hardin, then follows from a finite exact computation.

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1 The sequence and the conjecture

OEIS A275223 is "Number of nX3 0..2 arrays with no element equal to any value at offset (-2,-1) (-2,1) or (-1,0) and new values introduced in order 0..2.". It has offset 1 and begins

$$5, 36, 61, 187, 648, 2282, 8134, 29027, \ \ldots$$

The entry carries the following comment:

> Empirical: a(n) = 3*a(n-1) +5*a(n-2) -10*a(n-3) -a(n-4) -10*a(n-5) +21*a(n-6) -4*a(n-7) -4*a(n-8) +a(n-9) for n>13

As of the "Last modified" line on the live entry (Jul 20 2016 18:24:41, revision 4) the statement is still recorded as empirical, and nothing on the entry records it as settled either way.

Written out, the claim is

$$a(n) \;=\; 3\,a(n-1)+5\,a(n-2)-10\,a(n-3)-a(n-4)-10\,a(n-5)+21\,a(n-6)-4\,a(n-7)-4\,a(n-8)+a(n-9) \tag{1}$$

for all $n > 13$.

2 The condition, and what the canonical-form clause counts

The arrays have 3 columns and $L = n$ rows; write $x_{t,u}$ for the entry in row t and column u. With

$$\mathcal{N} = \{(-2,-1),\ (-2,1),\ (-1,0)\}$$

(offsets as (row shift, column shift)), the entry's requirement is

$$x_{t+d_1,\,u+d_2} \neq x_{t,u} \quad \text{for every } (d_1,d_2) \in \mathcal{N} \text{ inside the array.} \tag{2}$$

Every offset has a nonpositive row shift, so the condition at a cell of row t is settled by rows $t-2$, $t-1$ and t; but some shift is -2, so one previous row is not enough state.

The condition is stated purely in equalities, so it is unchanged by any relabelling of the values and is a property of the *equality pattern* of the array — the partition of the cells into classes of equal entries. The clause "new values introduced in order 0..2" keeps exactly one array from each relabelling class, so with $K=3$ values available the entry counts admissible patterns with at most K classes.

Lemma 1. *Let $L_i(n)$ count the arrays over an alphabet of i letters satisfying (2), with no canonical-form clause, and $N_j(n)$ the admissible patterns with exactly j classes. Then $L_i = \sum_j N_j\, i(i-1)\cdots(i-j+1)$ and*

$$3!\sum_{j\le 3} N_j(n) \;=\; \sum_{i=0}^{3}\binom{3}{i} D_{3-i}\, L_i(n), \qquad D_m = m!\sum_{t=0}^{m}\frac{(-1)^t}{t!}.$$

Proof. A labelled array satisfying the condition is a pattern together with an injection of its classes into the alphabet, and a pattern with j classes admits $i(i-1)\cdots(i-j+1)$ such injections; that is the first identity. Writing the falling factorial as $j!\binom{i}{j}$ gives $L_i=\sum_j (j!N_j)\binom{i}{j}$, so binomial inversion yields $j!N_j=\sum_i(-1)^{j-i}\binom{j}{i}L_i$. Summing over $j\le 3$ and exchanging the order of summation, the coefficient of L_i is $\sum_{t\le 3-i}(-1)^t/(i!\,t!)$, which on multiplying by $3!$ becomes $\binom{3}{i}(3-i)!\sum_{t\le 3-i}(-1)^t/t! = \binom{3}{i}D_{3-i}$, an integer. □

3 Each L_i is a walk count on pairs of rows

Fix i, let $V_i=\{0,\dots,i-1\}^3$ and take $\mathcal{V}=V_i\times V_i$ as vertices. Declare $(p,c)\to(c,x)$ an edge when every cell of the NEW row x satisfies (2), reading c as the row one step back and p as the row two steps back. Let P_i be the adjacency matrix and $\mathbf{s}_i$ the indicator of the pairs (p,c) in which p satisfies (2) with no row before it and c satisfies it with only p before it.

Lemma 2. *For $L\ge 2$ the arrays of L rows satisfying (2) are exactly the walks $(r^{(1)},r^{(2)})\to\cdots\to(r^{(L-1)},r^{(L)})$ of length $L-2$ starting at a vertex marked by $\mathbf{s}_i$, so $L_i=\mathbf{s}_i^\top P_i^{L-2}\mathbf{1}$.*

Proof. Consecutive vertices overlap in one row, so such a walk is precisely a sequence $r^{(1)},\dots,r^{(L)}$ of rows. The condition at the cells of row t for $t\ge 3$ is tested exactly once, at the step arriving at $(r^{(t-1)},r^{(t)})$, and with the correct earlier rows; the cells of rows 1 and 2 are tested by $\mathbf{s}_i$, with the rows that would lie before them absent, which is how the array behaves at its top. No further condition constrains the end of the array, so the walk may finish anywhere. □

Lemma 3. *Two vertices with the same equality pattern have, for each target pattern, the same number of successors carrying it, and the same value of $\mathbf{s}_i$. Hence the chain is strongly lumpable over patterns, and L_i may be computed on the quotient, whose states are the set partitions of the $2\cdot 3$ cells of a pair into at most i classes.*

Proof. Equal patterns means $(p',c')=(\sigma(p),\sigma(c))$ for a permutation σ of the alphabet. Condition (2) is a statement about equalities between entries, so it is invariant under σ; therefore $x\mapsto\sigma(x)$ is a bijection between the successor sets which preserves patterns, and $\mathbf{s}_i$ agrees on the two vertices. The number of pairs with pattern P is $e_P=i(i-1)\cdots(i-|P|+1)$, which supplies the weights. □

Assembling the 3 lumped chains block-diagonally into N of size $S=155$, with the weights of Lemma 1 folded into the start vector $\mathbf{v}$,

$$3!\;a(n) \;=\; \mathbf{v}^\top N^{\,L-2}\mathbf{1}, \qquad L=n. \tag{3}$$

4 The criterion, and the computation

Let $q(t) = t^9 - \sum_i c_i t^{9-i}$ be the characteristic polynomial of (1) and $G = N$.

Lemma 4. *With n_0 the least index for which $L \geq 2$ and $h_j = G^j N^o \mathbf{1}$, $o = 1n_0 + 0 - 2$, we have for $j \geq 9$*

$$3!\Big(a(n_0 + j) - \sum_i c_i\, a(n_0 + j - i)\Big) = \mathbf{v}^\top G^{j-9} w, \qquad w = h_9 - \sum_i c_i h_{9-i}.$$

Hence (1) holds from that point on if and only if $\mathbf{v}^\top G^m w = 0$ for every $m \geq 0$, and it is enough to check $m = 0, 1, \dots, S - 1$.

Proof. Substitute (3) and factor; the nonzero scalar 3! does not affect vanishing. By Cayley–Hamilton G^S is an integer combination of $I, G, \dots, G^{S-1}$, so each $\mathbf{v}^\top G^m w$ with $m \geq S$ is the corresponding combination of earlier values. □

Theorem 5. *The recurrence (1) holds for every $n > 13$, which is the statement of Section 1.*

Proof. The lumped transition counts were built by taking one representative pair for each pattern, enumerating every row x over the alphabet, testing (2) on x, and sorting the survivors by the pattern of the new pair. The vector w was formed by 9 applications of G in exact integer arithmetic and $\mathbf{v}^\top G^m w$ evaluated for $m = 0, 1, \dots,$ again exactly. Every one of those integers vanishes from the index corresponding to $n = 13 + 1$ onwards, and the run continued until $S + 1$ consecutive zeros had been seen. □

5 Verification

Three checks, each independent of the algebra above.

First, the right-hand side of (3), divided by 3!, was evaluated at every one of the 24 published indices and agrees with the entry's DATA exactly, the quotient coming out an integer each time; no shift was fitted, the number of rows being read off the entry's own name as $L = n$ and the entry's offset saying which index the first published term carries. A misreading of the offsets, of the inversion, or of the treatment of the first two rows would show up here.

Second, the lumped chain was checked against the unlumped one on the small shapes of this family, the walk counts over the full alphabet agreeing with the weighted sum over patterns term by term.

Third, the recurrence (1) was evaluated on the published terms in exact integer arithmetic with no matrices involved, and the annihilation test of Lemma 4 was carried to the full Cayley–Hamilton bound rather than to a sampled prefix.

References

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